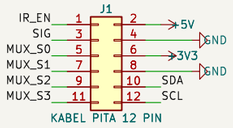
1. KE PCB UTAMA



KABEL PITA 12 PIN

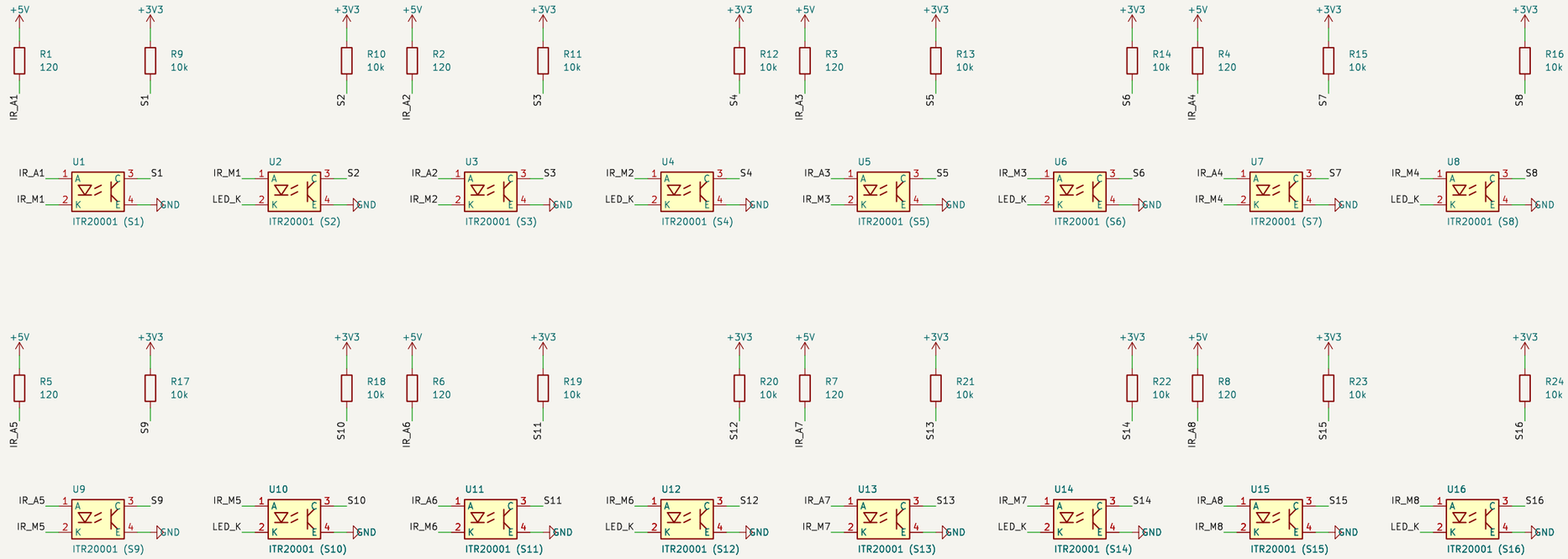
Kabel pita IDC 2x6 ke J5 PCB utama

PWR_FLAG (untuk ERC)



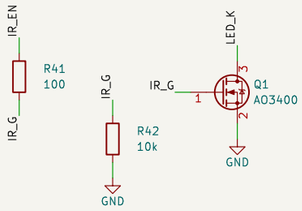
H1–H2: baut ke dudukan 3D print

2. SENSOR GARIS S1–S16 (ITR20001), LED IR dipasang seri berdua

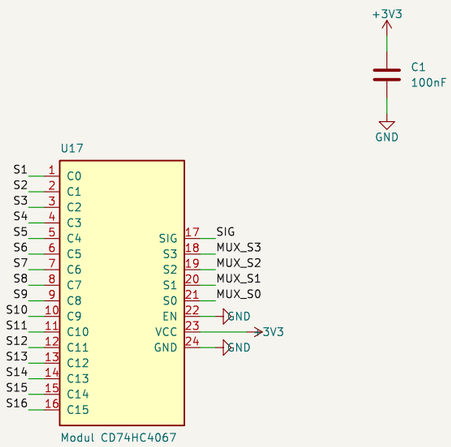


LED IR: (5 V − 2 x 1,2 V) / 120 = +−21 mA per pasang, total +−170 mA lewat Q1

3. SAKLAR LED IR

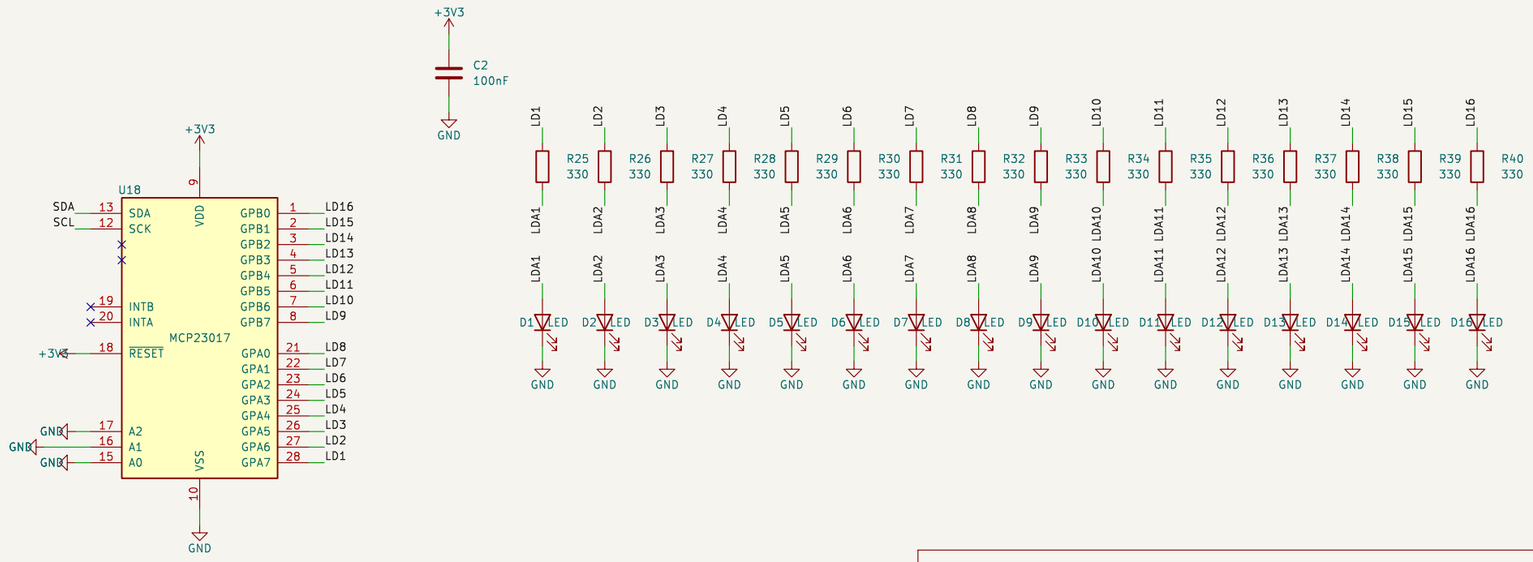


4. MULTIPLEXER 16 KE 1 (modul CD74HC4067)



Kanal C0–C15 = S1–S16 (C0 = S1). Modul dicolok di socket female

5. LED INDIKATOR TIAP SENSOR (MCP23017, I2C alamat 0x20)



LED1–8 = GPA7..GPA0, LED9–16 = GPB7..GPB0. MCP sumber arus: (3,3 − 1,9) / 330 = +−4 mA

v2 (13/10/2026): 16 sensor + mux CD74HC4067 + LED via MCP23017
S1 = paling kiri (dilihat dari atas, robot maju ke atas)
Acuan: docs/skematik−v1.md
PO1 Kel.5 PjBL DTEO FV ITS

Sheet: /	
File: ltm−sensor.kicad_sch	
Title: LTM − PCB Sensor 16 titik	
Size: A3	Date: 2026−10−13
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4	Id: 1/1

Rev: v2